

Hae Seong Lee

POSTDOCTORAL RESEARCHER IN STATISTICAL PHYSICS

Statistical Physics of Eco-Evolutionary Dynamics (SPEED) Lab, Department of Physics, Inha University

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Research Interests

- Emergent phenomena in complex systems
- Dynamics in complex networks
- Ecological Systems
- Synchronization

Research Experiences

Inha University

Incheon, South Korea

POSTDOCTORAL RESEARCHER

Mar. 2025 - Present

- Host: Prof. Hye Jin Park
- NRF postdoctoral fellow (Sep. 2026 – Aug. 2027)
- Random Lotka–Volterra dynamics on complex networks: how degree-correlated interaction strengths reshape biodiversity and network assortativity (PRE, under review)
- Eco-evolutionary model of migratory timing under seasonal resources and competition

Education

Sungkyunkwan University

Suwon, South Korea

PH.D. IN PHYSICS

Mar. 2019 - Feb. 2025

- Thesis: Clap, Sing, and Sleep: A Study of Emergent States in the Kuramoto Model
- Supervisor: Prof. Beom Jun Kim

Sungkyunkwan University

Suwon, South Korea

B.S. IN PHYSICS

Mar. 2013 - Aug. 2017

Publications

PUBLISHED

- [4] Younghwa Cha, Yeji Lee, Eunhee Ji, SoHyun Han, Sunhyun Min, Hyoungkyu Kim, Minseo Cho, **Hae Seong Lee**, Youngjai Park, and Joon-Young Moon, "Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI", Sci. Data (2026)
- [3] Jae Hyung Woo, **Hae Seong Lee**, Tae-Wook Ko and Joon-Young Moon, "Hysteresis in a generalized Kuramoto model with a first-order approximate coupling function and inhomogeneous coupling strengths", Chaos, Solitons and Fractals 190, 115770 (2025).
- [2] **Hae Seong Lee**, Beom Jun Kim, and Hye Jin Park, "Stability of twisted states in power-law-coupled Kuramoto oscillators on a circle with and without time delay", Phys. Rev. E 109, 064203 (2024).
- [1] **Hae Seong Lee**, Jong Il Park, and Beom Jun Kim, "Modified Kuramoto model with inverse-square law coupling and spatial time delay", Physica A 582, 126263 (2021).

PREPRINTS

- (submitted) **Hae Seong Lee**, Deok-Sun Lee, Sang Hoon Lee, Samir Suweis, and Hye Jin Park,

"How Degree-correlated Interaction Strength Modulates Degree Assortativity in Ecological Networks", arXiv:2607.18809.
IN PREPARATION

- **Hae Seong Lee** and Beom Jun Kim, "The effect of prior knowledge on vocabulary learning"
- **Hae Seong Lee**, Hyeong-Chai Jeong, Kalle Parvinen, Ulf Dieckmann, and Hye Jin Park, "Migration Patterns of Migratory Birds Driven by Seasonal Resource Variation and Competition"

Grants and Fellowships

Postdoctoral Research Fellowship

*National Research Foundation
of Korea (NRF)
Sep. 2026 - Aug. 2027*

- Principal Investigator. Project: Power Grid Synchronization Stability under Renewable Energy Expansion: An Approach Based on the Diversity-Stability Paradox

Honors and Awards

SKKU Scholarship Innovation

*Sungkyunkwan University
2022*

SKKU Innovative Research Fellowship

*Sungkyunkwan University
2022*

Excellence Project Award

*19th KIAS-APCTP Winter School
on Statistical Physics
2022*

Outstanding Oral Presentation Award

*2021 Korean Physical Society
Spring Meeting
2021*

Graduate Merit Scholarship

*Sungkyunkwan University
2019-2021*

Skills

Languages	Korean(native), English(fluent)
Programming Languages	C, C++, Python, SQL, Matlab, Javascript
Computer Skills	Monte-carlo simulation, CUDA programming, Agent-based simulation, MPI programming, Networkx, Latex

Experience

TEACHING ASSISTANT

Thermal and Statistical Mechanics II

SUNGKYUNKWAN UNIVERSITY

*Suwon, South Korea
Spring 2020, Spring 2021*

Thermal and Statistical Mechanics I

SUNGKYUNKWAN UNIVERSITY

*Suwon, South Korea
Fall 2020, Fall 2021*

Analytical Mechanics II Exercise

SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

Fall 2019

Analytical Mechanics II

SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

Fall 2019

Analytical Mechanics I

SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

Spring 2019

MILITARY SERVICE

Technical Research Personnel

SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

Mar. 2022 - Feb. 2025

Services

Organizer of the joint journal club of statistical physicists

SUNGKYUNKWAN UNIVERSITY, HANYANG UNIVERSITY, CATHOLIC UNIVERSITY OF KOREA, AND
INHA UNIVERSITY

South Korea

Aug. 2024 - Feb. 2025

Attended Schools

INTERNATIONAL SCHOOLS

Unifying Mathematical Models of Biodiversity

TOPIC: A THEORETICAL OVERVIEW OF ECOLOGICAL DYNAMICS AND COMPLEXITY

Tarascon-sur-Ariege, France

Jul. 2026

XII GEFENOL Summer School on Statistical Physics of Complex Systems

TOPIC: SELECTED TOPICS IN STATISTICAL PHYSICS

Madrid, Spain

Jul. 2024

Complexity72h

TOPIC: UNRAVELING CANCER DYNAMICS: FROM MULTISCALE STOCHASTIC MODELS TO TISSUE
MORPHOLOGY

Madrid, Spain

Jun. 2024

Complex Networks: Theory, Methods, and Applications

TOPIC: SELECTED TOPICS IN COMPLEX NETWORKS

Como, Italy

May 2024

CSH Winter School

TOPIC: INTEGRATIVE AND DISINTEGRATIVE PROCESSES IN COMPLEX HUMAN SOCIETIES

Obergurgl, Austria

Jan. 2023

VII Mediterranean school on complex networks

TOPIC: SELECTED TOPICS IN COMPLEX NETWORKS

Catania, Italy

Jun. 2022

DOMESTIC SCHOOLS

22nd KIAS-APCTP Winter School on Statistical Physics

TOPIC: STATISTICAL PHYSICS OF BIOLOGICAL AND SOFT MATTER

Pohang, South Korea

Jan. 2025

21st KIAS-APCTP Winter School on Statistical Physics

TOPIC: SELECTED TOPICS IN STATISTICAL PHYSICS ON COMPLEX SYSTEMS

Pohang, South Korea

Jan. 2024

Complex systems summer school

TOPIC: LECTURES ON "NETWORK SCIENCE" WRITTEN BY BARABASI

Seoul, South Korea

Jul. 2023

20th KIAS-APCTP Winter School on Statistical Physics

TOPIC: STATISTICAL PHYSICS OF QUANTUM SYSTEMS

Pohang, South Korea

Jan. 2023

19th KIAS-APCTP Winter School on Statistical Physics

TOPIC: PHASE TRANSITIONS AND CRITICAL PHENOMENA

Online

Jan. 2022

18th KIAS-APCTP Winter School on Statistical Physics

TOPIC: KINETIC MODELS IN STATISTICAL PHYSICS

Online

Jan. 2021

17th KIAS-APCTP Winter School on Statistical Physics

TOPIC: MACHINE LEARNING FOR STATISTICAL PHYSICS IN PRACTICE

Pohang, South Korea

Jan. 2020

16th KIAS-APCTP Winter School on Statistical Physics

TOPIC: CRITICAL PHENOMENA AND RENORMALIZATION GROUP

Yeosu, South Korea

Jan. 2019

Presentations

INTERNATIONAL ORAL PRESENTATION

- [8] "Species Survival Probability Under Degree-Interaction Strength Correlation", Arctic Meeting for Adaptive Mechanisms in Biological Systems (Abisko, Sweden, Jan. 20, 2026)
- [7] "Bird Migration Patterns Arising from Seasonal Resource Availability and Competition", Conference on Complex Systems 2025 (Siena, Italy, Sep. 1, 2025)
- [6] "Hysteresis in coupled identical oscillators with generalized coupling function and coupling strength inhomogeneity", Complex Networks 2023 (Menton, France, Nov. 28, 2023)
- [5] "Stability of twisted states in Kuramoto oscillators on a circle with power-law interaction strength", Roles of heterogeneity in non-equilibrium collective dynamics (RHINO 2022) (Tokyo, Japan, Sep. 18, 2022)
- [4] "Twisted states in Kuramoto oscillators on a circle with distance-decaying coupling strength", 15th Asia Pacific Physics Conference (Online, Aug. 22, 2022)
- [3] "Synchronization of oscillators with power-law coupling and spatial time delay", 2022 Korea-Canada Symposium and International Workshop on Multiplex Brain Networks (Calgary & Banff, Canada, Apr. 21, 2022)
- [2] "The central word set in a language", Networks 2021 (Online, Jul. 5, 2021)
- [1] "Modified Kuramoto model with power law coupling and spatial time delay", Conference on Complex Systems 2019 (Singapore, Sep. 30, 2019)

DOMESTIC ORAL PRESENTATION

- [9] "Degree-correlated Interactions Drive Assortativity in Ecological Networks through Hub-Favored and Hub-Suppressed Phases", 2026 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 23, 2026)
- [8] "Evolution of risk-taking strategy in von Neumann poker game", Workshop on the game theory and its applications (Gwangmyeong, South Korea, Nov. 24, 2023)
- [7] "Stability of Twisted States in Kuramoto Oscillators on a Circle with Distance-Decaying and Time-Delayed Coupling", 2023 Korean Physical Society Spring Meeting (Daejeon, Apr. 20, 2023)
- [6] "The compartment learning model and the core vocabulary in Korean language", 2021 Korean Physical Society Spring Meeting (Online, Apr. 22, 2021)
- [5] "A set of central words in Korean language", 2020 Korea Academy of Complexity Studies Fall Conference (Online, Nov. 28, 2020)
- [4] "A set of central words in Korean language", Joint mini-workshop on collective dynamics (Busan, South Korea, Nov. 12, 2020)
- [3] "A set of central words in Korean language", 2020 Korean Physical Society Fall Meeting (Online, Nov. 5, 2020)
- [2] "Modified Kuramoto model with power law coupling and site-dependent time delay", 2019 Korean Physical Society Fall Meeting (Gwangju, South Korea, Oct. 24, 2019)
- [1] "Modified Kuramoto model with power law coupling and spatial time delay", The 20th Workshop for Statistical Physics (Byeonsan, South Korea, Aug. 21, 2019)

INTERNATIONAL POSTER PRESENTATION ---

- [6] "How Degree-Dependent Interactions Reshape Hub Survival in Ecological Networks", Netsci 2026 (Boston, USA, Jun. 3, 2026)
- [5] "Generalized Random Lotka-Volterra model with degree-correlated interaction strengths", Statphys29 (Florence, Italy, Jul. 14, 2025)
- [4] "How Do Degree-Correlated Interaction Strengths Shape Biodiversity?", ACMB-JSMB 2025 (Kyoto, Japan, Jul. 8, 2025)
- [3] "Why we sing in a round at a stadium: stability analysis of twisted states in Kuramoto oscillators on a circle", Netsci 2024 (Quebec, Canada, Jun. 17, 2024)
- [2] "Finding core vocabulary in a language", Statphys28 (Tokyo, Japan, Aug. 8, 2023)
- [1] "Effect of prior knowledge on vocabulary learning", Conference on Complex Systems 2022 (Palma, Spain, Oct. 18, 2022)

DOMESTIC POSTER PRESENTATION ---

- [5] "Impact of Degree-Correlated Interactions on Biodiversity", 2025 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 23, 2025)
- [4] "Biodiversity in Random Lotka-Volterra model with degree-correlated interaction strength", Stochastic Processes as Tools for Understanding Biological Systems (Busan, South Korea, Feb. 17, 2025)
- [3] "Locally twisted states of Kuramoto oscillators in a brain network with site-dependent time delay", 2024 Korean Physical Society Fall Meeting (Yeosu, South Korea, Oct. 25, 2024)
- [2] "Finding important words: Network analysis of a Korean dictionary", 2024 Korean Physical Society Spring Meeting (Daejeon, South Korea, Apr. 25, 2024).
- [1] "Emergence of twisted states in Kuramoto oscillators on a circle with power-law coupling strength", 2022 SKKU Physics Workshop (Suwon, South Korea, Nov. 30, 2022)